AGVAN GRIGORYAN

Systems Software Engineer

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Detail-oriented systems engineer with a strong foundation in C, POSIX APIs, and Linux architecture. Experienced in writing robust OS-level code, handling concurrent processing, and engineering automated Linux infrastructure. Highly capable of navigating low-level memory management and utilizing OS debugging tools. Integrates AI coding assistants (Claude, Gemini) critically to accelerate script generation and troubleshooting while strictly maintaining code security and binary stability. Seeking to apply system-level debugging and C programming skills to vulnerability patching and extended lifecycle support.

CORE TECHNICAL SKILLS

Systems Programming	C, C++, POSIX APIs, Multithreading (pthreads), Memory Management, GCC/Clang, Make
Linux / OS Administration	Debian/Ubuntu/CentOS, Bash Scripting, systemd, journald, networking, POSIX environments
Debugging & Quality	strace, gdb, valgrind, memory leak analysis, code auditing, unit testing pipelines
Infrastructure & Tooling	Git, GitHub Actions, Docker, Vagrant, AI-Assisted Engineering (Gemini, Claude)

SYSTEMS & ALGORITHMS PROJECTS

Minishell (Unix Shell Implementation) — 42 Yerevan

C, POSIX API, Bash internals, valgrind

- Developed a fully functional Unix shell from scratch, accurately replicating core bash execution logic.
- Interacted directly with OS system calls (`fork`, `execve`, `pipe`, `dup2`, `waitpid`) to manage multi-piped command execution and I/O redirection.
- Implemented complex signal handling using `sigaction` to gracefully manage interrupts (ctrl-C, ctrl-D, ctrl-\) without crashing running background processes.

Push_swap (Algorithmic Optimization) — 42 Yerevan

C, Data Structures, Complexity Analysis

- Engineered a highly optimized sorting algorithm ("Butterfly" strategy) using only two stacks and a restricted instruction set.
- Implemented custom doubly linked lists to achieve `O(1)` time complexity for stack rotations and push/pop operations, completely avoiding costly array memory shifts.
- Developed a robust input parser capable of validating edge cases such as integer overflow/underflow, duplicate values, and non-numeric tokens.

Philosophers (Concurrency & Multithreading) — 42 Yerevan

C, pthreads, Mutex locks, Concurrency

- Solved the classic Dining Philosophers problem by designing a highly optimized multithreaded C simulation.
- Utilized POSIX threads to manage concurrent execution states and implemented fine-grained mutex locks to protect shared resources, successfully preventing data races and deadlocks.

Automated Linux Infrastructure (k3s & SysAdmin) — 42 Yerevan

Debian/CentOS, Bash, systemd, Virtualization

- Authored idempotent Bash scripts to fully provision and configure multi-node Linux virtual machine clusters from a blank OS installation.
- Configured logical volume management (LVM) and hardened OS security by enforcing strict UFW firewall policies and customizing the SSH daemon.
- Diagnosed complex system-level failures—including control-plane resource exhaustion and swap-induced daemon crashes—using `journalctl` and `systemctl`.

PROFESSIONAL EXPERIENCE

Backend Developer & Systems Administrator — NIVI (nivi.ge)

01/2025 – Present | Yerevan, Armenia

- Architected and deployed a containerized production platform, managing the entire lifecycle from backend code to Linux server provisioning.
- Automated background job processing and maintenance tasks using scheduled queues and cron operations.

Python Developer (Traineeship) — Andersen

01/2026 – 04/2026 | Yerevan, Armenia

- Developed containerized backend services communicating via REST APIs within a microservices architecture.

EDUCATION

42 Yerevan (part of TUMO Labs) — 2024 – Present

Computer Science & Systems Engineering

Rigorous, peer-to-peer project-based curriculum heavily focused on C programming, memory management, algorithms, and Unix system architecture.

Yerevan State College of Informatics — 2019 – 2023

Software Engineering Diploma